

This assignment will not be graded and is only for practice.

2.1 Level 1:

1) Suppose $A = \begin{bmatrix} 2 & 4 & 5 & 1 \\ 11 & -2 & 9 & -6 \\ -3 & 4 & 7 & 7 \end{bmatrix}$. Which of the following is true about the matrix **1 point**

A ?

[Hint: The (i, j) -th entry is the entry which is at the i -th row and j -th column.]

- ☐ It is a 4×3 matrix.
- ☒ It is a 3×4 matrix.
- ☐ $(2, 3)$ -th entry of the matrix A is 4.
- ☒ $(2, 3)$ -th entry of the matrix A is 9.

Yes, the answer is correct.

Score: 1

Accepted Answers:

It is a 3×4 matrix.

$(2, 3)$ -th entry of the matrix A is 9.

2) Which of the following statements is(are) TRUE? **1 point**

[Hint: Recall, the definitions of scalar matrix, diagonal matrix, and identity matrix.]

- ☐ Any diagonal matrix is a scalar matrix.
- ☐ Scalar matrices may not be square matrices.
- ☒ Scalar matrices must be square matrices.
- ☐ Any scalar matrix is an identity matrix.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Scalar matrices must be square matrices.

3) **1 point**

Given below is the system of linear equations:

$$7x_1 + 10x_2 + 12x_3 = 36$$

$$8x_1 + 4x_2 - 9x_3 = 11$$

$$4x_1 - x_2 + 3x_3 = 10$$

If the matrix representation of the system of equations is $Ax = b$, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, then

- ☒ $A = \begin{bmatrix} 7 & 10 & 12 \\ 8 & 4 & -9 \\ 4 & -1 & 3 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$
- ☐ $A = \begin{bmatrix} 7 & 10 & 12 & 36 \\ 8 & 4 & 9 & 11 \\ 4 & 1 & 3 & 10 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$
- ☐ $A = \begin{bmatrix} 7 & 10 & 12 & 36 \\ 8 & 4 & -9 & 11 \\ 4 & -1 & 3 & 10 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$
- ☐ $A = \begin{bmatrix} 7 & 10 & 36 & 12 \\ 8 & 4 & 11 & -9 \\ 4 & -1 & 10 & 3 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$$A = \begin{bmatrix} 7 & 10 & 12 \\ 8 & 4 & -9 \\ 4 & -1 & 3 \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix}$$

4) Which of the following statements is(are) TRUE?

1 point

- ☐ Addition of two matrices is possible only if the number of columns in the first matrix is same as the number of rows in the second matrix.
- ☒ Addition of two matrices is possible only if the orders of both the matrices are the same.
- ☐ Defining AB is possible if the number of rows in the matrix A is same as the number of columns in the matrix B .

- ☒ Defining AB is possible if the number of columns in the matrix A is same as the number of rows in the matrix B .

Yes, the answer is correct.

Score: 1

Accepted Answers:

Addition of two matrices is possible only if the orders of both the matrices are the same.
Defining AB is possible if the number of columns in the matrix A is same as the number of rows in the matrix B .

5) Suppose $P = \begin{bmatrix} 3 & -1 & 7 \\ 4 & 0 & 1 \\ 2 & -5 & 2 \end{bmatrix}$, $Q = [1 \quad 4 \quad -9]$, $R = [0 \quad -3 \quad 10]$, $D = \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix}$ **1 point**

[Hint: If A is a matrix of order $m \times n$ and B is a matrix of order $n \times p$, then the order of AB is $m \times p$.]

- ☒ The matrix PD is of order 3×1 .
- ☐ The matrix PD is of order 1×3 .
- ☐ The matrix QD is of order 3×3 .
- ☒ The matrix QD is of order 1×1 .
- ☒ The matrix DQ is of order 3×3 .
- ☐ The matrix DQ is of order 1×1 .
- ☐ The product QD is not defined.
- ☒ The product QR is not defined.
- ☒ The addition $P + Q$ is not defined.
- ☒ The addition $P + D$ is not defined.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The matrix PD is of order 3×1 .

The matrix QD is of order 1×1 .
The matrix DQ is of order 3×3 .
The product QR is not defined.
The addition $P + Q$ is not defined.
The addition $P + D$ is not defined.

2.2 Level 2:

6) Choose the set of correct options.

1 point

[Hint: Assume $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and use the given conditions to find matrix A , for first two options.]

- ☐ If A is a square matrix of order 2 and $A^2 = I$, then $A = I$ or $A = -I$ where I is the identity matrix of order 2.
- ☐ If A is a square matrix of order 2 and $A^2 = 0$, then $A = 0$.
- ☒ If A and B are square matrices of order 3 and $A + B = 0$, then $B = -A$.
- ☒ If A is a scalar matrix of order 3, B is a non-zero square matrix of order 3 and $AB = 0$, then $A = 0$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

If A and B are square matrices of order 3 and $A + B = 0$, then $B = -A$.

If A is a scalar matrix of order 3, B is a non-zero square matrix of order 3 and $AB = 0$, then $A = 0$.

7) If A is a square matrix of order 2 whose first column is denoted by C_1 and second column is denoted by C_2 (the left most column is taken as the first one) and $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$ then choose the set of correct options.

1 point

[Hint: Recall the definition of matrix multiplication in terms of rows and columns.]

- ☒ The first column of AB is $b_{11}C_1 + b_{21}C_2$.
- ☒ The second column of AB is $b_{12}C_1 + b_{22}C_2$.
- ☐ The first column of AB is $b_{21}C_1 + b_{11}C_2$.
- ☐ The second column of AB is $b_{22}C_1 + b_{21}C_2$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The first column of AB is $b_{11}C_1 + b_{21}C_2$.

The second column of AB is $b_{12}C_1 + b_{22}C_2$.

8) Choose the set of correct options.

1 point

[Hint: Assume $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and use the given condition to find matrix A for last two options.]

- ☒ There exist some real matrices A and B , such that $AB = BA$.
- ☐ There do not exist any real matrices A and B , such that $AB = BA$.
- ☐ There does not exist any real matrix A , such that $A^2 = A$.
- ☒ There exists some real 2×2 matrix A , such that $A^2 + A + I = 0$

Yes, the answer is correct.

Score: 1

Accepted Answers:

There exist some real matrices A and B , such that $AB = BA$.

There exists some real 2×2 matrix A , such that $A^2 + A + I = 0$

9) Suppose A is a 3×3 scalar matrix and (1,1)-th entry of the matrix A is 4. Suppose B is a 3×3 square matrix such that (i,j) -th entry is equal to $i^2 + j^2$. Find the (2,2)-th entry of the matrix $2A + B$.

16

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 16

1 point

10) Suppose $A = \begin{bmatrix} -1 & 2 \\ -4 & 7 \end{bmatrix}$ and $A^2 - \alpha A + I = 0$ for some $\alpha \in \mathbb{R}$.

Find the value of α .

6

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 6

1 point

Your score is: 10/10